



BACKLOG SUBJECT EXAMINATIONS - JULY / AUGUST 2025

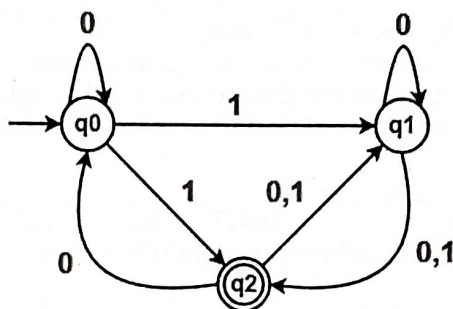
B.E. - CSE (Cyber Security) /			
Program	: CSE (Artificial Intelligence and Machine Learning)	Semester	: V
Course Name	: Automata Theory and Compiler Design	Max. Marks	: 100
Course Code	: 21 / 22 / CY53 / CI53	Duration	: 3 Hrs.

Instructions to the Candidates:

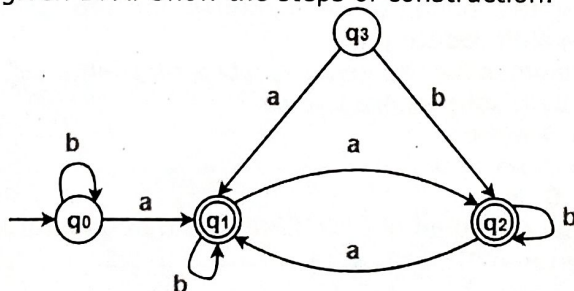
- Answer one full question from each unit.

UNIT - I

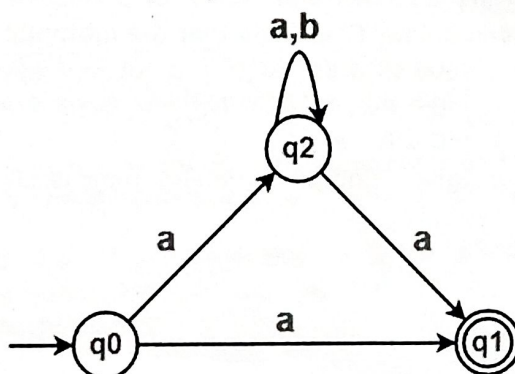
- Illustrate the frontend and backend of the compiler by translating the assignment statement. Assume all variables are integers.
Res=val+val*10+sum; CO4 (08)
 - Construct a DFA that accepts a language L over input alphabets $\Sigma = \{a, b\}$ such that L is the set of all strings starting with 'aa' or 'bb'. CO1 (06)
 - Convert the given NFA to DFA. CO4 (06)



- Minimize the given DFA. Show the steps of construction. CO4 (07)



- Convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA). Construct the Transition Table. CO4 (07)



- c) Describe the components for constructing a NFA. Construct an NFA with $\Sigma = \{0, 1\}$ that accepts all strings that begin with 1 and end with 0. CO4 (06)

UNIT - II

3. a) "Assume that all our Transition diagrams are deterministic". Justify your answer for this statement. CO1 (08)
Construct the transition diagram for the given tokens
i. logical operators
ii. while, which, what, wait
- b) Describe the languages denoted by the following regular expressions CO2 (06)
i. $(a|b)^*a?(a|b)$
ii. $a^*b+a^*b+a^+$
- c) Identify the token, pattern and lexeme for the given code fragment. CO2 (06)

```
int fact(int n){
    int f;
    if (n==1)
        return f;
}
```
4. a) Illustrate the role of lexical analyzer and interaction with parser with a neat diagram. State the tasks of lexical analyzer in the compiler design process. CO2 (09)
- b) Write character class for the following: CO2 (05)
i. Except All uppercase and lowercase vowels
ii. Strings containing the characters 'a', 'b', 'c' and 'd' in order.
- c) Write the formal definition of Pushdown Automata. Specify the graphical notation of PDA's. CO1 (06)

UNIT - III

5. a) Construct LR(0) parse table for the Grammar Given CO3 (09)
 $G: S \rightarrow L$
 $L \rightarrow (L)L \mid a \mid \epsilon$
 Do parsing for the input: (a)\$
- b) Illustrate how the handles are identified for the given grammar by constructing a shift reduce parser. CO3 (06)
Identify the handles for the given inputs string "abbcd" by showing the moves made by a shift reduce parser.
 $G: S \rightarrow aABe$
 $A \rightarrow Abc \mid b$
 $B \rightarrow d$
- c) Give the formal definition of CFG. Give CFG that generates the strings for the Language $L = \{0^n1^n \mid n \geq 1\}$. CO3 (05)
6. a) Consider the grammar given below. If not suitable for Predictive Parsing make necessary changes and construct predictive parsing table for the grammar given below. Check whether the grammar is LL(1). CO3 (08)
 $G: A \rightarrow Bd \mid a$
 $B \rightarrow Ae \mid b \mid \epsilon$
 $C \rightarrow B \mid e$
- b) Prove that the given Context Free Grammar is LR(1) by generating the parse table. CO3 (06)
 $S \rightarrow AaAb \mid BbBa$
 $A \rightarrow \epsilon$
 $B \rightarrow \epsilon$

- c) Consider the grammar $E \rightarrow E+E \mid E * E \mid E-E \mid x \mid y$. Find leftmost and rightmost derivation and construct corresponding parse trees for the string $x+y * x-y$. CO3 (06)

UNIT- IV

7. a) Describe the language of a Turing Machine. Specify the Turing Machine and halting problem. CO4 (07)
 b) Explain the rules for turning an L-attributed SDD to SDT. Convert the SDD given below to SDT. CO5 (07)

Production	Semantic Rules
$D \rightarrow B D'$	$D.val = D'.syn$ $D'.inh = B.val$
$D' \rightarrow B D_1'$	$D_1'.inh = D'.inh * 2 + B.val$ $D'.syn = D_1'.syn$
$D' \rightarrow \epsilon$	$D'.syn = D'.inh$
$B \rightarrow 0$	$B.val = 0$
$B \rightarrow 1$	$B.val = 1$

- c) Write S-attributed Definition for the given CFG to count the number of bits in the given binary value '1011'. Draw an annotated parse tree for the same. CO5 (06)
 G: $N \rightarrow D$
 $D \rightarrow D_1 B \mid B$
 $B \rightarrow 0 \mid 1$
8. a) Explain the Semantic Rules with Controlled Side Effects by generating an SDD for basic type declarations Grammar: $D \rightarrow T L$
 $T \rightarrow \text{int} \mid \text{float}$
 $L \rightarrow L_1, \text{id} \mid \text{id}$
 Draw the dependency graph for the following expression
 int a, b, c CO5 (06)

- b) Consider the following attribute grammar: CO3 (06)

Production	Semantic Rule
$R \rightarrow D S$	$S.u = D.v$
$S \rightarrow A B C$	$B.u = S.u$ $A.u = B.v + C.v$ $S.v = A.v$
$A \rightarrow a$	$A.v = 2 * A.u$
$B \rightarrow b$	$B.v = B.u$
$C \rightarrow c$	$C.v = 1$
$D \rightarrow 3$	$D.v = 3$

- Draw the parse tree for the input 'abc', (the only string for the language), and show the dependency graph for the associated attributes.
- Describe one correct order for the evaluation of the attributes. Assume 'u' is the inherited attribute and 'v' is the synthesized attribute.
- Assume that S.u is assigned the value of 3 before starting attribute evaluation, What will be the value of S.v when evaluation has terminated.

- c) Write a L-Attributed SDD for used for Simple desk calculator. Obtain the Semantic action, Annotated parse tree and dependency Graph for the string $3*5+4n$. CO5 (08)

G: $E \rightarrow TE'$
 $E' \rightarrow +TE' \mid \epsilon$
 $T \rightarrow FT'$
 $T' \rightarrow FT' \mid \epsilon$
 $F \rightarrow \text{digit}$

UNIT – V

9. a) Translate the following control flow statement with the help of semantic actions of one pass code generation. CO5 (08)
 $\text{if}(a < b) \ a = 1; \text{ else } a = 0;$
- b) Construct 3AC and Annotated Parse Tree for CO5 (06)
i) $x = a[i] + b[j]$ ii) $x = c + a[1][2]$
- c) Determine the types and relative address for the following CO5 (06)
 $\text{float } x;$
 $\text{record}\{$
 $\text{int } y;$
 $\text{int } z;$
 $\}\text{q};$
10. a) Construct DAG and generate its corresponding three address code for the following expression. CO5 (09)
 $a = a + b + (a + b)$
Draw the Syntax Tree and generate its equivalent three address code for the same expression. Does both generates the same three address code sequence? If not, state the reason.
- b) Write the three address code and quadruple representation for the given statements CO5 (06)
i. $f = \min(1, n-1, n+1)$
ii. $a = x[i] + b * c$
- c) Explain the implementation of getReg(I) function, for a three-address instruction, such as $I: a = b + c$. CO4 (05)
